Lab 08: Java Generics, Comparable interface, Collection Framework

Exercise 01:

1. Create an abstract class named `LibraryItem` with the following protected attributes:
   protected String title;
   protected String author;
   protected int year;
2. Define a constructor that initializes all three attributes.
3. Declare the following abstract method: public abstract void displayInfo();
4. Implement the `Comparable<LibraryItem>` interface by overriding `compareTo()` to sort items **by year** in ascending order.
5. Create two concrete subclasses: `Book` and `Magazine`, each extending `LibraryItem`.
6. Override `displayInfo()` in both subclasses to print relevant item details.
7. Create a custom exception class named `DuplicateItemException` that accepts a descriptive message upon instantiation.
8. Create a `Library` class containing a private `ArrayList<LibraryItem>` named `items`.
9. Implement `addItem()` — if an item with the same title already exists in the list, throw a `DuplicateItemException`.
10. Implement `displayAllItems()` to iterate over and display all stored items.
11. Implement `sortItems()` using: Collections.sort(items);
12. In your `main` method:
    - Instantiate several `Book` and `Magazine` objects
    - Add them to the library inside a `try-catch` block to handle `DuplicateItemException`
    - Display all items **before** sorting
    - Call `sortItems()`
    - Display all items **after** sorting

Exercise 02:

1. Create a generic class named `StorageBox<T>` with a single private attribute `item` of type `T`.
2. Implement `setItem()` to assign a value to `item`.
3. Implement `getItem()` to return the stored value.
4. Implement `displayItem()` to print the stored value to the console.
5. In your `main` method, create and demonstrate the following instances:
   - A `StorageBox<String>`
   - A `StorageBox<Integer>`
   - A `StorageBox<Book>` (reusing the `Book` class from Exercise 1)
6. Call `displayItem()` on each box to verify its stored content.